

Anonymous Supplementary Technical Report for SimP

Abstract

This supplementary technical report provides detailed per-benchmark baseline comparisons and ablation results accompanying an anonymous SimP submission. The material is separated from the main paper and is intended solely to provide additional experimental detail during review.

1 Detailed Baseline Comparisons

Table 1 reports percentage changes in reduction time and final program size for the baseline comparisons. It uses all available cases for each baseline: 44 for LPR and 44 for the other configurations.

Table 1: Percentage changes in reduction time and final program size for SimP relative to each baseline. Entries express geometric-mean SimP/baseline ratios as percentage changes. Negative values indicate less time or fewer tokens; positive values indicate more.

Baseline	C (23)		Rust (11)		JavaScript (10)		All (44)	
	Time (%)	Size (%)	Time (%)	Size (%)	Time (%)	Size (%)	Time (%)	Size (%)
Perses + T-Rec	-25.8	-44.0	-15.2	-12.6	+62.9	-30.5	-8.3	-34.3
+ Latra	-35.3	-20.6	-18.8	-12.6	+61.2	-31.3	-15.7	-21.3
LPR	-55.4	+10.8	-73.8	+4.1	+2.9	-10.4	-52.8	+4.0
C-Reduce	-36.4	+36.0	-53.0	-10.6	-71.6	-17.0	-50.9	+9.5

Per-benchmark comparisons. Table 2 shows results across the 44 benchmarks for all five configurations, SimP is the fastest configuration on 22 cases and achieves the smallest output, including ties, on 16 cases. The fastest baseline varies across inputs: selecting it separately for each benchmark yields performance on par with SimP in terms of geometric-mean runtime ($1.00\times$), compared with $2.12\times$ against the fixed LPR baseline. The two dominated cases are `gcc118638` and `js-4`. On the former, + Latra produces 147 tokens in 2,343 seconds versus SimP’s 230 tokens in 3,315 seconds. On the latter, LPR produces 33 tokens in 100 seconds versus SimP’s 42 tokens in 171 seconds.

2 Ablation Results by Language

Table 3 provides the language breakdown for the ablation study. It uses the same 44 benchmarks, expressing the variant/full SimP ratios as percentage changes. **Per-benchmark comparisons.** Table 4 shows results across the 44 benchmarks for all five ablation configurations. SimP achieves the smallest output, including ties, on 42 cases, while the variant without Mutator is the fastest on all 44 cases. Removing Mutator reduces the runtime to $0.72\times$ that of SimP in geometric mean, but increases the output size to $1.97\times$. SimP is Pareto-dominated on only two cases: `gcc110115` and `rust-63791`. On the former, the naive-prompt variant produces the same 228 tokens in 11,842 seconds, compared with SimP’s 12,183 seconds. On the latter, the variant without Mutator produces 3,454 tokens in 2,422 seconds, compared with SimP’s 4,195 tokens in 5,670 seconds.

Table 2: Comparison of CReduce, Perses, Latra, LPR, and SimP on gcc-, Rust, and JavaScript benchmarks in terms of reduction time, final token count, and monetary cost. T-Rec is enabled by default in Perses, Latra and LPR.

Bug	O (#)	CReduce		Perses		Latra		LPR			SimP			Best Approach	
		T (s)	R (#)	T (s)	R (#)	T (s)	R (#)	T (s)	R (#)	C (\$)	T (s)	R (#)	C (\$)	smallest	fastest
gcc-106892	31979	8137	203	13503	425	14906	292	25754	208	0.255	6869	199	0.374	SimP	SimP
gcc-107176	51016	2684	84	2436	190	2621	135	3485	110	0.164	1349	62	0.286	SimP	SimP
gcc-107407	31670	4724	103	4395	250	5006	156	5459	92	0.191	1849	103	0.313	LPR	SimP
gcc-109778	54844	2408	104	1439	300	1597	189	3776	151	0.205	1757	102	0.340	SimP	Perses
gcc-109925	31109	6355	136	4513	346	4845	221	5821	181	0.215	2256	267	0.357	CReduce	SimP
gcc-109967	55526	5307	114	2857	296	3146	196	5610	189	0.234	2886	151	0.327	CReduce	Perses
gcc-110115	105468	23019	78	8517	321	9159	276	15363	218	0.227	12183	228	0.355	CReduce	Perses
gcc-110141	68129	7582	181	6917	374	7949	333	14578	254	0.293	4028	243	0.357	CReduce	SimP
gcc-110243	40714	1999	128	2305	294	2595	229	6042	155	0.268	1518	184	0.417	CReduce	SimP
gcc-110640	47833	4790	142	4680	314	5097	248	7311	191	0.309	3548	232	0.427	CReduce	SimP
gcc-111137	34540	3169	125	2324	257	2740	163	2740	127	0.231	1682	133	0.288	CReduce	SimP
gcc-111331	58245	2578	86	1801	255	1852	147	4360	76	0.139	2539	118	0.327	LPR	Perses
gcc-111445	51091	7199	101	5561	319	10801	239	9874	128	0.125	3663	139	0.367	CReduce	SimP
gcc-111472	19032	2703	107	2113	336	2333	311	4469	163	0.194	1802	140	0.355	CReduce	SimP
gcc-111583	51059	4148	160	2064	422	2350	284	6560	231	0.300	3779	245	0.414	CReduce	Perses
gcc-111764	24199	2571	106	2503	185	3050	61	2628	42	0.106	1458	69	0.338	LPR	SimP
gcc-116906	253235	65579	61	61901	274	63364	229	48732	115	0.140	24523	260	0.425	CReduce	SimP
gcc-117424	35021	4761	148	10392	378	17296	251	12651	140	0.225	3582	268	0.338	LPR	SimP
gcc-117498	107896	11540	167	11894	386	13435	298	31035	415	0.255	9174	291	0.339	CReduce	SimP
gcc-117574	58217	11605	88	9710	127	10555	105	7812	96	0.212	2938	72	0.340	SimP	SimP
gcc-118623	43104	20306	118	21869	204	21929	204	25551	137	0.142	13198	141	0.345	CReduce	SimP
gcc-118638	50545	3053	106	2295	226	2343	147	5163	132	0.167	3315	230	0.455	CReduce	Perses
gcc-118953	29980	1448	122	1612	243	1801	159	3384	80	0.109	3279	114	0.372	LPR	CReduce
rust-44800	801	17610	471	2177	424	2261	424	9658	129	0.220	1199	325	0.496	LPR	SimP
rust-63791	8143	16673	4562	13735	4563	18285	4560	124741	4594	0.185	5670	4195	1.438	SimP	SimP
rust-66851	936	2557	649	6127	713	6132	713	9877	713	0.229	3906	639	0.436	SimP	CReduce
rust-69039	190	2307	109	1610	114	1611	114	4631	109	0.222	494	99	0.499	SimP	SimP
rust-77002	347	3676	264	266	247	269	247	1028	247	0.176	431	211	0.328	SimP	Perses
rust-91725	513	861	101	358	114	379	114	1092	59	0.184	279	83	0.269	LPR	SimP
rust-99830	448	1746	164	1074	303	1079	303	2642	294	0.204	1151	262	0.260	CReduce	Perses
rust-111502	192	258	161	71	161	77	161	1480	161	0.188	257	132	0.284	SimP	Perses
rust-112061	556	1917	447	2014	447	2015	447	2621	447	0.183	1999	436	0.425	SimP	CReduce
rust-112213	866	1692	726	2310	643	2312	643	4134	643	0.190	1652	721	0.446	Perses	SimP
rust-112526	644	1718	537	2096	382	2097	382	3290	382	0.183	2220	330	0.400	SimP	CReduce
js-1	244	345	41	135	52	134	52	200	35	0.018	224	31	0.229	SimP	Latra
js-2	112	1074	40	196	51	194	52	266	41	0.021	247	41	0.155	CReduce	Latra
js-3	125	835	47	152	57	153	57	279	38	0.034	244	47	0.207	LPR	Perses
js-4	185	1084	47	55	59	57	52	100	33	0.018	171	42	0.216	LPR	Perses
js-5	178	737	51	79	57	80	47	121	47	0.013	210	39	0.250	SimP	Perses
js-6	152	1416	45	289	40	289	55	309	39	0.009	238	19	0.199	SimP	SimP
js-7	144	841	38	55	46	54	46	93	44	0.012	132	39	0.149	CReduce	Latra
js-8	121	741	45	96	55	100	55	313	34	0.072	165	42	0.200	LPR	Perses
js-9	87	392	32	157	44	165	44	221	36	0.014	188	32	0.172	CReduce	Perses
js-10	63	334	43	176	51	175	56	204	51	0.011	205	31	0.156	SimP	Latra

Columns O , T , and R list the number of original tokens, the reduction time, and the number of tokens after reduction, respectively.

Table 3: Percentage changes in reduction time and final program size for each ablation variant relative to full SimP. Entries express geometric-mean variant/SimP ratios as percentage changes. Negative values indicate less time or fewer tokens; positive values indicate more.

Variant	C (23)		Rust (11)		JavaScript (10)		All (44)	
	Time (%)	Size (%)	Time (%)	Size (%)	Time (%)	Size (%)	Time (%)	Size (%)
Without reducer	-6.9	+88.7	-8.4	+43.1	-2.9	+118.0	-6.4	+81.9
Without mutator	-26.6	+98.4	-40.7	+55.1	-15.9	+159.1	-28.2	+98.2
Without Perses	-19.3	+68.8	-24.2	+40.7	-8.7	+104.7	-18.3	+68.5
Naive prompt	+3.8	+33.7	+3.0	+14.7	+1.0	+34.4	+3.0	+28.8

Table 4: Ablation study of SimP on GCC, Rust, and JavaScript benchmarks in terms of reduction time and final token count.

Bug	O (#)	Without Reducer		Without Mutator		Without Perses		SimP (Naive Prompt)		SimP	
		T (s)	R (#)	T (s)	R (#)	T (s)	R (#)	T (s)	R (#)	T (s)	R (#)
gcc-106892	31979	5985	415	4671	280	4913	249	7599	295	6869	199
gcc-107176	51016	1370	191	908	115	1084	80	1321	82	1349	62
gcc-107407	31670	1802	278	1283	162	1401	152	2184	155	1849	103
gcc-109778	54844	2013	535	1246	382	1646	283	1800	146	1757	102
gcc-109925	31109	2778	576	927	930	1090	845	2728	289	2256	267
gcc-109967	55526	2654	309	2200	238	2357	199	3003	240	2886	151
gcc-110115	105468	11484	271	11080	352	11222	305	11842	228	12183	228
gcc-110141	68129	3497	269	3128	363	3312	274	4062	273	4028	243
gcc-110243	40714	1247	281	953	283	1080	271	1653	247	1518	184
gcc-110640	47833	2960	493	2237	849	2482	735	2937	247	3548	232
gcc-111137	34540	1511	275	1130	188	1266	192	1925	148	1682	133
gcc-111331	58245	2550	209	2126	330	2267	246	2585	173	2539	118
gcc-111445	51091	3337	234	3043	222	3233	146	3817	204	3663	139
gcc-111472	19032	1586	238	1321	433	1524	326	1860	194	1802	140
gcc-111583	51059	3509	457	3130	745	3358	653	4025	294	3779	245
gcc-111764	24199	1288	121	1088	166	1305	113	1782	154	1458	69
gcc-116906	253235	23243	438	20826	739	21211	684	25110	300	24523	260
gcc-117424	35021	2397	434	1697	569	1855	491	3649	370	3582	268
gcc-117498	107896	8825	444	8273	487	8583	429	9302	294	9174	291
gcc-117574	58217	2663	155	2168	127	2441	104	2668	264	2938	72
gcc-118623	43104	12965	328	12177	176	12395	153	13485	139	13198	141
gcc-118638	50545	3231	278	2870	265	3173	351	3370	264	3315	230
gcc-118953	29980	3233	169	2955	198	3112	183	3278	131	3279	114
rust-44800	801	1095	458	701	494	903	446	1236	370	1199	325
rust-63791	8143	4912	4430	2422	4562	3615	5020	5952	4375	5670	4195
rust-66851	936	3427	789	1799	826	2592	775	4083	692	3906	639
rust-69039	190	465	176	345	200	409	168	504	123	494	99
rust-77002	347	408	324	310	356	362	313	439	248	431	211
rust-91725	513	268	153	219	175	245	146	283	104	279	83
rust-99830	448	1053	385	679	419	870	373	1186	303	1151	262
rust-111502	192	247	222	205	249	228	213	260	161	257	132
rust-112061	556	1794	580	1054	618	1425	567	2073	485	1999	436
rust-112213	866	1493	869	905	905	1202	856	1710	774	1652	721
rust-112526	644	1986	464	1146	500	1565	451	2305	375	2220	330
js-1	244	217	69	184	83	202	65	227	42	224	31
js-2	112	238	87	199	102	220	82	250	54	247	41
js-3	125	235	97	197	114	218	91	247	62	244	47
js-4	185	167	89	148	104	159	83	172	56	171	42
js-5	178	204	83	175	99	190	78	212	52	210	39
js-6	152	230	47	193	57	213	44	241	27	238	19
js-7	144	130	83	121	99	126	78	133	52	132	39
js-8	121	161	89	144	104	154	83	166	56	165	42
js-9	87	183	71	160	85	173	67	190	43	188	32
js-10	63	199	69	171	83	186	65	207	42	205	31

O , T and R denote the number of original tokens, reduction time, and number of tokens after reduction, respectively.